Model Question Paper Set - 1 with effect from 2022(CBCS Scheme)

USN | | | | | | | | | | |

Fourth Semester B.E Degree Examination

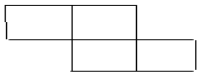
DISCRETE MATHEMATICAL STRUCTURES (BCS405A)

TIME: 03Hours **Max.Marks:100**

Note:

1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
2. M: Marks, L: RBT levels, C: Course outcomes.

		Module – 1	M	L	C
Q.1	a	Show that the compound proposition $\left[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)\right] \Leftrightarrow \left[(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)\right]$ for primitive statements p, q, r is logically equivalence.	6	L2	CO1
	b	Establish the validity of the following argument using the Rules of Inference: $\{p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \sim q)\} \rightarrow (s \vee t)$.	7	L2	CO1
	c	For the universe of all integers, let p(x), q(x), r(x), s(x) and t(x) denote the following open statements: p(x): x>0, q(x): x is even, r(x): x is a perfect square, s(x): x is divisible by 3, t(x): x is divisible by 7. Write the following statements in symbolic form: i) At least one integer is even. ii) There exists a positive integer that is even. iii) If x is even, then x is not divisible by 3. iv) No even integer is divisible by 7. v) There exists even integer divisible by 3.	7	L1	CO1
		OR			
Q.2	a	Define a tautology. Prove that, for any propositions p, q, r the compound propositions, $\{(p \rightarrow q) \wedge (q \rightarrow r)\} \rightarrow \{(p \rightarrow r)\}$ is tautology.	6	L2	CO1
	b	Prove the following using laws of logic: $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$.	7	L2	CO1
	c	Give i) direct proof ii) indirect proof iii) proof by contradiction for the following statement: "if n is an odd integer then n+9 is an even integer".	7	L3	CO1
		Module – 2			
Q.3	a	Prove that $1^2 + 3^2 + 5^2 + \cdots . + (2n-1)^2 = \frac{n(2n+1)(2n-1)}{3}$ by Mathematical Induction.	6	L2	CO2
	b	Let $a_0 = 1, a_1 = 2, a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$. Prove that $a_n \leq 3^n \ \forall \ n \in z^+$.	7	L2	CO2
	c	Find the number of ways of arrangement of the letters of the word 'TALLAHASSEE' which have no adjacent A's.	7	L2	CO2
		OR			
Q.4	a	Determine the coefficient of xyz^2 in the expansion of $(2x - y - z)^4$.	6	L2	CO2
	b	In how many ways one can distribute 8 identical marbles in 4 distinct containers so that i) no container is empty ii) the fourth container has an odd number of marbles in it.	7	L2	CO2
	c	How many positive integers n can we form using the digits 3,4,4,5,5,6,7 if we want n to exceed 5,000,000?	7	L2	CO2
		Module – 3			
Q.5	a	Let f: R → R be defined by, $f(x) = \begin{cases} 3x - 5 \ if \ x > 0 \\ 1 - 3x \ if \ x \leq 0 \end{cases}$. Find $f^{-1}(0), f^{-1}(1), f^{-1}(-1), f^{-1}(3), f^{-1}(6), \ f^{-1}([-6,5])$ and $f^{-1}([-5,5])$	6	L2	CO3

	b	State Pigeon hole principle. Prove that in any set of 29 persons; at least 5 persons have been born on the same day of the week.	7	L2	CO3
	c	Let A={1,2,3,4,6} and 'R' be a relation on 'A' defined by aRb if and only if "a is multiple of b" represent the relation 'R' as a matrix, draw the diagraph and relation R.	7	L2	CO3
		OR			
Q.6	a	If $f:A\rightarrow B$, $g:B\rightarrow C$, $h:C\rightarrow D$ are three functions, then Prove that h∘(g∘f)=(h∘g)∘f.	6	L2	CO3
	b	Show that if n+1 numbers are chosen from 1 to 2n then at least one pair add to 2n+1.	7	L2	CO3
	c	Draw the Hasse diagram representing the positive divisors of 72.	7	L2	CO3
		Module – 4			
Q.7	a	In how many ways the 26 letters of English alphabet are permuted so that none of the pattern's CAR, DOG, PUN or BYTE occurs?	6	L2	CO4
	b	Define Derangement. In how many ways can each of 10 people select a left glove and a right glove out of a total of 10 pairs of gloves so that no person selects a matching pair of gloves?	7	L2	CO4
	c	Solve the recurrence relation: $C_n = 3C_{n-1} - 2C_{n-2}, for\ n \geq 2$, given $C_1 = 5, C_2 = 3$.	7	L3	CO4
		OR			
Q.8	a	In how many ways one can arrange the letters of the word **CORRESPONDENTS** so that there are i) exactly 2 pairs of consecutive identical letters? ii) at least 3 pairs of consecutive identical letters? iii) no pair of consecutive identical letters?	6	L2	CO4
	b	Find the rook polynomial for the chess board as shown in the figure 	7	L2	CO4
	c	Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0, a_0 = 1, a_1 = 6$.	7	L3	CO4
		Module – 5			
Q.9	a	If H, K are subgroups of a group G, prove that $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G.	6	L2	CO5
	b	Define Klein 4 group. Verify $A = \{1, 3, 5, 7\}$ is a klein 4 group.	7	L2	CO5
	c	State and prove Lagrange's Theorem.	7	L2	CO5
		OR			
Q.10	a	Show that i) the identity of G is unique. ii) the inverse of each element of G is Unique.	6	L3	CO5
	b	Show that $(A, \cdot)$ is an abelian group where $A = \{a \in Q \mid a \neq -1\}$ and for any $a, b \in A, a \,.\, b = a + b + ab$.	7	L3	CO5
	c	Let $G = S_4$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup $H = \langle \alpha \rangle$. Determine the left cosets of H in G.	7	L3	CO5

Model Question Paper Set - 2 with effect from 2022(CBCS Scheme)

USN | | | | | | | | | | |

Fourth Semester B.E Degree Examination

DISCRETE MATHEMATICAL STRUCTURES (BCS405A)

TIME:03Hours **Max.Marks:100**

Note:

1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
2. M: Marks, L: RBT levels, C: Course outcomes.

		Module - 1	M	L	C
Q.1	a	Define tautology. Show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology by constructing the truth table.	6	L1	CO1
	b	Prove the following using the laws of logic $[\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)] \Leftrightarrow r$.	7	L2	CO1
	c	For any two odd integers m and n, show that (i) m+n is even (ii) mn is odd.	7	L2	CO1
		OR			
Q.2	a	Define i) an open statement ii) Quantifiers	6	L2	CO1
	b	Write the following argument in symbolic form and then establish the validity If A gets the Supervisor's position and works hard, then he will get a raise. If he gets a raise, then he will buy a car. He has not purchased a car. Therefore he did not get the Supervisor's position or he did not work hard.	7	L1	CO1
	c	For the following statements, the universe comprises all non-zero integers. Determine the truth value of each statement. a) $\exists\, x \exists\, y\, [xy = 1]$ b) $\exists\, x \forall y\, [xy = 1]$ c) $\forall x \exists\, y\, [xy = 1]$ d) $\exists\, x \exists\, y [(2x + y = 5) \wedge (x - 3y = -8)]$ e) $\exists\, x \exists\, y [(3x - y = 7) \wedge (2x + 4y = 3)]$	7	L2	CO1
		Module – 2			
Q.3	a	Define the well ordering principle. By Mathematical Induction, Prove that $(\boldsymbol{n}!) \geq \boldsymbol{2n-1}$ for all integers $\boldsymbol{n} \geq \boldsymbol{1}$.	6	L2	CO2
	b	Prove that every positive integer n≥24 can be written as a sum of 5's and/or 7's.	7	L3	CO2
	c	How many positive integers n , can we form using the digits 3,4,4,5,5,6,7, if we want n to exceed 5,000,000.	7	L1	CO2
		OR			
Q.4	a	By Mathematical Induction Prove that $1.3 + 2.4 + \cdots \ldots \ldots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$.	6	L1	CO2
	b	Find the number of permutations of the letters of the word MASSASAUGA. In how many of these all four *A's* are together? How many of them begin with *S*?	7	L2	CO2
	c	i) Obtain the Coefficient of a^5b^2 in the expansion of $(2a-3b)^7$ ii) Using the Binomial theorem find the coefficient of x^5y^2 in	7	L1	CO2

		the expansion of $(x+y)^7$.			
		Module – 3			
Q.5	**a**	State Pigeon –hole principle. Prove that if any number from1 to 8 are chosen then two of them will have their sum as 9.	**6**	**L1**	**CO3**
	b	Let $f: R \to R$ be defined by ,$f(x) = \begin{cases} 3x-5, & if\ x > 0 \\ 1-3x, & if x \le 0 \end{cases}$ $find, f^{-1}([-6,5])\ and\ f^{-1}([-5,5])$.	**7**	**L1**	**CO3**
	c	Let $A = B = C = R$, and $f: A \to B$ and $g: B \to C$ be defined by $f(a) = 2a+1, \quad g(b) = \frac{1}{3}b, \forall a \in A, \forall b \in B.$ Compute gof and show that $gof$is invertible. What is $(g\ o\ f)^{-1}$?	**7**	**L2**	**CO3**
		OR			
Q.6	**a**	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$, If $(g\ o\ f)(x) = 9x^2 - 9x + 3$ determine a and b.	**6**	**L3**	**CO3**
	b	Draw the Hasse (POSET) diagram which represents positive divisors of 36.	**7**	**L2**	**CO3**
	c	Let $A = \{1,2,3,4,6\}$ and R be a relation on A defined by aRb if and only if "a is a multiple of b". Write down the relation R, relation matrix $M(R)$ And draw its diagraph. List out its in degree and out degree.	**7**	**L2**	**CO3**
		Module – 4			
Q.7	**a**	Determine the number of positive integers n such that $1 \le n \le 100$ and n is not divisible by 2, 3, or 5	**6**	**L2**	**CO4**
	b	In how many ways can the 26 letters of the English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?	**7**	**L2**	**CO4**
	c	Solve the recurrence relation $a_n = na_{n-1}$ where n ≥ 1and $a_0 = 1$.	**7**	**L2**	**CO4**
		OR			
Q.8	**a**	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block?	**6**	**L3**	**CO4**
	b	Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5, one teacher for each class. $T_1$and T_2 do not wish to become the class teachers for C_1or C_2, T_3 and T_4 for C_4 or C_5 , and $T_5$for C_3 or C_4 or C_5 .In how many ways can the teachers be assigned the work (without displeasing any teacher	**7**	**L2**	**CO4**
	c	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$where n ≥ 0 and $F_0 = 0, F_1 = 1$.	**7**	**L2**	**CO4**
		Module – 5			
Q.9	**a**	Define group. Show that fourth roots of unity is an abelian group.	**6**	**L2**	**CO5**
	b	If G be a set of all non-zero real numbers and let a* b = ab/2 then show that (G,*) is an abelian group.	**7**	**L2**	**CO5**
	c	Define Klein 4-group. And if A = { e,a,b,c} then show that this is a Klein -4 group	**7**	**L1**	**CO5**
		OR			
Q.10	**a**	Define Cyclic group and show that (G,8) whose multiplication table is as given below is Cyclic	**6**	**L2**	**CO5**

			*	a	b	c	d	e	f			
			a	a	b	c	d	e	f			
			b	b	c	d	e	f	a			
			c	c	d	e	f	a	b			
			d	d	e	f	a	b	c			
			e	e	f	a	b	c	d			
			f	f	a	b	c	d	e			
	b	State and prove Lagrange's theorem								**7**	**L1**	**CO5**
	c	If G be a group with subgroup H and K. If $\|G\|$ = 660 and $\|K\|$ = 66 and K C H C G and find the possible value for $\|H\|$								**7**	**L2**	**CO5**

:- Model Question Paper-1&2 :-

:- Discrete Mathematical Structures [BCS405A] :-

:- Module-1 :-

① Show that the Compound proposition $[(P \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow P)] \Leftrightarrow [(P \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow P)]$ for primitive Statements P, q, r is logically equivalence.

→

P	q	r	$P \leftrightarrow q$	$q \leftrightarrow r$	$r \leftrightarrow P$	$[(P \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow P)]$
0	0	0	1	1	1	1
0	0	1	1	0	0	0
0	1	0	0	0	1	0
0	1	1	0	1	0	0
1	0	0	0	1	0	0
1	0	1	0	0	1	0
1	1	0	1	0	0	0
1	1	1	1	1	1	1

$P \rightarrow q$	$q \rightarrow r$	$r \rightarrow P$	$[(P \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow P)] \Leftrightarrow [(P \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow P)]$
1	1	1	1
1	1	0	1
1	0	1	1
1	1	0	1
0	1	1	1
0	1	1	1
1	0	1	1
1	1	1	1

∴ P, q, r is logically equivalence

(b) Establish the validity of the following argument using Rules of inference: $\{P \wedge (P \rightarrow q) \wedge (S \vee r) \wedge (r \rightarrow \neg q)\} \rightarrow (S \vee t)$

→ Premises : $P \wedge (P \rightarrow q) \wedge (S \vee r) \wedge (r \rightarrow \neg q)$

Conclusion : $S \vee t$.

P
P→q
S∨u
u→~q

∴ from P & P→q by modus ponens

(i) P, P→q / q

u→~q, q / ~u

∴ from q & u→~q by Modus Tollens

∴ from S∨u & ~u by disjunctive Syllogism,

~u, ~S∨u / S

we have S, by addn (a tautology in classical logic) we can write S∨t

∴ It is valid.

(c) For the universe of all integers, let P(x), q(x), u(x), s(x), & t(x) denote the following Open Statements: P(x): x>0, q(x): x is even, u(x): x is Perfect Square, s(x): x is divisible by 3, t(x): x is divisible by 7. Write the following Statements in Symbolic form:

(i) At least One integer is even → ∃ x∈S, q(x)

(ii) There exists a +ve integer that is even
→ ∃ x∈S, [P(x) ∧ q(x)]

(iii) If x is even, then x is not divisible by 3
→ q(x) → ~s(x)

(iv) No even integer is divisible by 7 → ~t(x)

(v) There exists even integer divisible by 3
→ ∃x∈S, [q(x) ∧ s(x)]

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(2)(a) Define Tautology. Prove that, for any Propositions P, q, u the Compound propositions {(P→q)∧(q→u)} → {(P→u)} is To.

→ Compound Propositions which are always True.

P	q	u	(P→q)	{(q→u)	{(P→q)∧(q→u)}	(P→u)
0	0	0	1	1	1	1
0	0	1	1	1	1	1
0	1	0	1	0	0	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0

1	0	1	0	1	0	1
1	1	0	1	0	0	0
1	1	1	1	1	1	1

$\{(P \to q) \wedge (q \to r)\} \longrightarrow \{(P \to r)\}$
1
1
1
1
1
1
1
1

∴ given Compound proposition is tautology.

(b) P.T using laws of logic : $P \to (q \to r) \Leftrightarrow (p \wedge q) \to r$.

→ $P \to (q \to r) \Leftrightarrow (p \wedge q) \to r$

LHS : $P \to (q \to r)$ $[P \to q = \sim P \vee q]$ implication

$\sim P \vee (q \to r)$

$\sim P \vee (\sim q \vee r)$ [implication]

$(\sim P \vee \sim q) \vee r$ [Associative & Commutative]

$\sim (p \wedge q) \vee r$ [Demorgans laws]

$(p \wedge q) \to r$ [Implication]

RHS : $(p \wedge q) \to r \Rightarrow P \to (q \to r)$

$(p \wedge q) \to r$

$\sim (p \wedge q) \vee r$ [Implication]

$(\sim P \vee \sim q) \vee r$ [Demorgans law]

$\sim P \vee (\sim q \vee r)$ [Associative law]

$\sim P \vee (q \to r)$ [Implication]

$P \to (q \to r)$ [Implication]

$(p \wedge q) \to r \Rightarrow P \to (q \to r)$

∴ $P \to (q \to r) \Rightarrow (p \wedge q) \to r$ and

$(p \wedge q) \to r \Rightarrow P \to (q \to r)$

∴ $P \to (q \to r) \Leftrightarrow (p \wedge q) \to r$.

(c) Give (i) Direct proof (ii) Indirect proof (iii) Proof by Contradiction for following statement : "if n is odd integer then n+9 is even integer".

→ p : n is odd int q : n+9 is an even integer.

(i) Direct proof : S.T p→q is true

Hypothesis : Assume that p is true i.e n is odd

Analysis : $n = 2k+1$

$n+9 = 2k+1+9$

$n+9 = 2k+10$, divisible by 2.

∴ n+9 is even integer ∴ q is true

Conclusion : p→q is true.

(ii) Indirect proof : ST ~q→~p is true

Hypothesis : Assume that ~q is true i.e n+9 is odd int

Analysis : $n+9 = 2k+1$

$n = 2k+1-9 = 2k-8$ is divisible by 2.

∴ n is even int i.e ~p is true.

Conclusion : ~q→~p is true. p→q is true.

(iii) Proof by Contradiction :

Hypothesis : Assume p→q is false i.e p is true and q is true.

Analysis : Take p is true by using rules, laws and any other known facts to S.T q is false. Our assumption is wrong.

Conclusion : p→q is false.

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(7) (a) Define Tautology. ST $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is tautology by Constructing truth table.

→ Compound propositions which are always true are called Tautology.

P	q	r	p∨q	p→r	q→r	[(p∨q)∧(p→r)∧(q→r)]
0	0	0	0	1	1	0
0	0	1	0	1	1	0
0	1	0	1	1	0	0
0	1	1	1	1	1	1
1	0	0	1	0	1	0
1	0	1	1	1	1	1
1	1	0	1	0	0	0
1	1	1	1	1	1	1

[(p∨q)∧(p→r)∧(q→r)] → r
1
1
1
1
1
1
1
1
1

∴ [(p∨q)∧(p→r)∧(q→r)] → r is always true which proves it is tautology.

(b) P.T following using laws of logic

[~p∧(~q∧r)]∨[(q∧r)∨(p∧r)] ⇔ r

→ [~p∧(~q∧r)]∨[(q∧r)∨(p∧r)]

⇔ [(~p∧~q)∧r]∨(q∧r)∨(p∧r) [Associative law]

⇔ [~(p∨q)∧r]∨(q∧r)∨(p∧r) [Demorgans law]

⇔ [~(p∨q)∧r]∨[(q∨p)∧r] [Distributive law]

⇔ [~(p∨q)∨(q∨p)]∧r

⇔ [~(p∨q)∨(p∨q)]∧r [Commutative law]

⇔ T₀∧r [Complement law]

⇔ r [Identity law]

(c) For any two Odd integers m and n, Show that

(i) m+n is even.

→ Any Odd integer can be written in the form $2k+1$, where k is an integer.

Let m & n be two Odd integers

$m = 2a+1$ & $n = 2b+1$, a & b are integers

$m+n = (2a+1)+(2b+1) = 2a+2b+2$

$m+n = 2(a+b+1)$

$(a+b+1)$ is int, $m+n$ is of form $2k$ where $2k$ is an integer.

∴ $m+n$ is even.

(ii) mn is Odd

→ $m = 2a+1$ $n = 2b+1$

$mn = (2a+1)(2b+1) = 2a.2b+2a+2b+1$

$mn = 4ab+2a+2b+1$

$mn = 2(2ab+a+b)+1$

Since $(2ab+a+b)$ is int, mn as $2 \times$ (an int) $+1$ which is an Odd integer

∴ mn is Odd.

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(2) (a) Define

(i) Open Statement :- is a logical Statement that Conta-ains one (or) more variables & becomes a proposi-tion (true (or) false) when specific values are substi-tuted for these variables.

Ex :- $P(x) : x+5 > 10$

The Statement is Open bcz truth value depends on value of x.

(ii) Quantifiers :- are words refer to Quantities like such as Some (or) All.

(b) Write the following argument in Symbolic form & then establish the validity. If A gets the Supervisor's position & works hard, then he will get a raise. If he get a raise, then he will buy a Car. He has not purchased a Car. Therefore he did not get the Supervisor's Position (or) he did not work hard.

→ P: A gets the Supervisor's position
Q: A Works hard
R: He will get a raise & S: He will buy a Car

$(P \wedge Q) \rightarrow R$
$R \rightarrow S$
$\sim S$

$\sim P \vee \sim Q$

$\Leftrightarrow (P \wedge Q) \rightarrow R \wedge (R \rightarrow S) \wedge \sim S \Rightarrow \sim P \vee \sim Q$

$(P \wedge Q) \rightarrow R \wedge [R \rightarrow S \wedge \sim S]$ [By Modus Tollens]

$(P \wedge Q) \rightarrow R \wedge \sim R$

$\sim (P \wedge Q)$ [By Modus Tollens]

$\sim P \vee \sim Q$ [De Morgans law]

$\sim P \vee \sim Q \Rightarrow [(P \wedge Q) \rightarrow R \wedge R \rightarrow S \wedge \sim S]$

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(5) For the following Statements, the universe Comprises all non-zero int. Determine the truth value of each Statement.

(a) $\exists x \exists y [xy = 1]$

→ This Statement claims that there exist non-zero int x & y so that their product is 1.

Ex:- $x = 1$ & $y = 1$ $(1 \times 1) = 1$

$x = -1$ & $y = -1$ $(-1) \times (-1) = 1$

∴ Since there are such pairs (x, y) the Statement is true.

(b) $\exists x \forall y [xy = 1]$

→ This Statement claims that there exist non-zero int x such that for all non-zero int y the product is 1.

* If x were fixed non-zero int, then $y = 1/x$ to be true for all y, but y is restricted to non-zero int & $1/x$ may not be an int for all x.

Ex:- $x = 2$ $y = 1/2$ is not an integer.

∴ Such an x does not exist, making statement false.

(c) $\forall x \exists y [xy = 1]$

→ This statement claims that for every non-zero int x, there exists an non-zero int y such that product is 1.

* For any non-int x, $y = 1/x$. $1/x$ must be integer.
* The only pairs (x, y) that satisfy $xy = 1$, where both x & y are int are (1, 1) & (-1, -1)
* $y = 1/x$ does not always result in int for every integer x, the statement is false.

(d) $\exists x \exists y [(2x + y = 5) \wedge (x - 3y = -8)]$

→ This statement claims that there exist non-zero int x & y that satisfy both equations.

$2x + y = 5$ $\qquad 2x + y = 5 \rightarrow (1)$

$x - 3y = -8 \times 2 \Rightarrow \quad 2x - 6y = -16 \rightarrow (2)$

Subtract second eqn from first eqn.

$(2x + y) - (2x - 6y) = 5 - (-16)$

$2x + y - 2x + 6y = 5 + 16$

$7y = 21 \quad \boxed{y = 3}$

Substitute $y = 3$ in eqn (1)

$2x + 3 = 5 \Rightarrow 2x = 2 \quad \boxed{x = 1}$

This statement is true, $x = 1$ & $y = 3$ are non-zero integer that satisfy both equation.

(e) $\exists x \exists y [(3x - y = 7) \wedge (2x + 4y = 3)]$

→ This statement claims that there exist non zero int x & y that satisfy both eqn simultaneously

$3x - y = 7 \times 4 \Rightarrow 12x - 4y = 28 \rightarrow (1)$

$2x + 4y = 3 \qquad 2x + 4y = 3 \rightarrow (2)$

Add $12x - 4y + 2x + 4y = 28 + 3$

$14x = 31 \quad \boxed{x = 31/14}$

∴ x is not int, there are no soln to this sys of eqns.

Module 2

(5)(a) P.T $1^2+3^2+5^2+\cdots+(2n-1)^2 = \frac{n(2n+1)(2n-1)}{3}$ by M.I

→ Let S(n) denote the given Statement

Basic Step: We note that if S(1) is the Statement $1^2 = \frac{1}{3}\cdot 1\cdot(2-1)\cdot(2+1)$ which is clearly true.

Induction Step: We assume that the Statement S(n) is true for $n=k$ where k is an int ≥ 1; that is, we assume that the following Statement is true.

$$S(k) = 1^2+3^2+5^2+\cdots+(2k-1)^2 = \frac{k(2k+1)(2k-1)}{3}$$

Adding $(2k+1)^2$ to both sides of this, we obtain.

$$S(k) = 1^2+3^2+5^2+\cdots+(2k-1)^2+(2k+1)^2 = \frac{k(2k+1)(2k-1)}{3}+(2k+1)^2$$

$$= (2k+1)\left\{\frac{k(2k-1)}{3}+(2k+1)\right\}$$

$$\Rightarrow \frac{1}{3}\left((2k+1)\{k(2k-1)+3(2k+1)\}\right.$$

$$= \frac{1}{3}\left((2k+1)\{2k^2-k+6k+3\}\right.$$

$$= \frac{1}{3}\left((2k+1)\{2k^2+5k+3\}\right.$$

$$= \frac{1}{3}(2k+1)(k+2)(2k+3)$$

This is precisely the Statement S(k+1). Thus, On the basis of the assumption that S(n) is true for $n=k\geq 1$, the truthness of S(n) for $n=k+1$ is established.

(b) Let $a_0=1$, $a_1=2$, $a_2=3$ and $a_n = a_{n-1}+a_{n-2}+a_{n-3}$ for $n\geq 3$. Prove that $a_n \leq 3^n \;\forall\; n\in z^+$.

→ Basic Step: We first verify the base cases a_0, a_1, a_2:

① $a_0 = 1 \leq 3^0 = 1$

② $a_1 = 2 \leq 3^1 = 3$

③ $a_2 = 3 \leq 3^2 = 9$

∴ These basic cases holds true.

Induction Step: Assume that $a_k \leq 3^k$ for $k = n, n-1, n-2$

$a_{n-1} \leq 3^{n-1}$, $a_{n-2} \leq 3^{n-2}$, $a_{n-3} \leq 3^{n-3}$

We need to show that $a_n \leq 3^n$.

Given the recurrence relation.

$a_n = a_{n-1} + a_{n-2} + a_{n-3}$

Using the inductive hypothesis

$a_n \leq 3^{n-1} + 3^{n-2} + 3^{n-3}$

We need to show that:

$3^{n-1} + 3^{n-2} + 3^{n-3} \leq 3^n$

Factoring Out 3^{n-3} from the left hand side:

$3^{n-1} + 3^{n-2} + 3^{n-3} = 3^{n-3}(3^2 + 3 + 1) = 3^{n-3} \cdot 13$

We need to show that $3^{n-3} \cdot 13 \leq 3^n$

$13 \leq 3^3 \Rightarrow 13 \leq 27$

∴ This inequality is true, hence we have:

$3^{n-1} + 3^{n-2} + 3^{n-3} \leq 3^n$

Thus, $a_n \leq 3^n$ holds true for all $n \in Z^+$.

By M.I, we have proven that $a_n \leq 3^n$ for all positive integers n.

(c) Find the number of ways of arrangement of the letters of the word 'TALLAHASSEE' which have no adjacent A's.

→ The no of letters in the given word is 11 of which 3 are A's, 2 each are L's, S's and E's and 1 each are T & H.

∴ the no of arrangements (permutations) of the letters in the given word is

$$\frac{11!}{3!\,2!\,2!\,2!\,1!\,1!} = 831600$$

If we discarded the A's the remaining 8 letters can be arranged in

$$\frac{8!}{2!\,2!\,2!\,1!\,1!} = 5040$$

In each of these arrangements, there are 9 possible locations for the three A's. These locations can be chosen in $C(9,3)$ ways. The no of arrangements having no Adjacent A's is $5040 \times C(9,3) = 5040 \times \frac{9!}{3!\,6!}$

$= 5040 \times 84 = 423360$

(4)(a) Determine the Coefficient of xyz^2 in the expansion of $(2x-y-z)^4$.

→ The multinomial theorem states that

$$(a_1+a_2+\dots+a_m)^n = \sum_{k_1+k_2+\dots k_m=n} \frac{n!}{k_1!k_2!\dots k_m!} a_1^{k_1} a_2^{k_2} \dots a_m^{k_m}$$

We need to determine exponents k_1, k_2, k_3 for $2x, -y, -z$ such

(1) $k_1+k_2+k_3 = 4$ (the power of whole expansion)

(2) The resulting term Contains x, y, z^2

we identify k_1, k_2, k_3 as follows:

$2x$ must be raised to power of 1 to get x

$-y$ must be raised to power of 1 to get y

$-z$ must be raised to power of 2 to get z^2.

$k_1 = 1, k_2 = 1, k_3 = 2 \Rightarrow 1+1+2 = 4$

multinomial Coefficient: $\frac{4!}{1!1!2!}(2x)^1(-y)^1(-z)^2$

$$\frac{4!}{1!1!2!} = \frac{24}{1.1.2} = \frac{24}{2} = 12$$

$$= 12.(2x)^1.(-y)^1.(-z)^2 = 12.2x.(-y).z^2$$

$$= 12.2.x.(-y).z^2 = 24.(-1).x.y.z^2 = -24xyz^2$$

∴ The Coefficient of xyz^2 in the expansion of $(2x-y-z)^4$ is -24.

(b) In how many ways One can distribute 8 identical ma-rbles in 4 distinct Containers so that

(i) no Container is empty.

→ First we distribute One (marble) ball to each Container, then the remaining 4 (marbles) balls is to be distributed among 4 Containers and this Can be done in

$$C(4+4-1;4) = C(7,4) = 35 \text{ ways}.$$

(ii) The fourth Container has an Odd number of marbles in it.

→ If the 4th Container has to get an Odd no of marbles we have to put 1 (or) 3 (or) 5 (or) 7 marbles in it.

Case (i): Suppose we put 1 marble ball to 4th Container, the remaining 7 marbles can distributed into remaining 3 Containers in $C(3+7+1, 7) = C(9, 7)$ ways.

Case (ii): Suppose we put 3 marbles to 4th Container the remaining 5 marbles can be distributed into remaining 3 Containers in $C(3+5-1, 5) = C(7, 5)$ ways.

Case (iii) Suppose we put 5 marbles to 4th Container the remaining 3 marbles can be distributed into remaining 3 Containers in $C(3+3-1, 3) = C(5, 3)$ ways.

Case (iv) Suppose we put 7 marbles to 4th Container the remaining 1 marble can be distributed into remaining 3 Containers in $C(3+1-1, 1) = C(3, 1)$ ways.

∴ Total no of ways of distributing the given marbles with the given condition is

$C(9,7) + C(7,5) + C(5,3) + C(3,1) = 70$.

© How many positive integers n can we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want to n to exceed 5,000,000?

→ Let $n = x_1 x_2 x_3 x_4 x_5 x_6 x_7$

where $x_1, x_2 \ldots x_7$ are the given digits.

Since n exceeds 5,000,000, x_1 can take 5 (or) 6 (or) 7

If $x_1 = 5$ then $x_2 \ldots x_7$ is an arrangement of remaining 6 digits which contains 2 4's & one each are 3, 5, 6, 7

∴ No of such arrangements is $= \frac{6!}{2!1!1!1!1!} = 360$.

If $x_1 = 6$ then $x_2 \ldots x_7$ is an arrangement of remaining 6 digits which contains 2 4's & 2 5's & one each are 3, 7

∴ No of such arrangements is $= \frac{6!}{2!2!1!1!} = 180$.

If $x_1 = 7$, then $x_2 \ldots x_7$ is an arrangement of the remaining 6 digits which contains 2 4's, 2 5's, 1 each of 3, 6.

∴ No of such arrangements is $= \frac{6!}{2!2!1!1!} = 180$

∴ By Sum Rule, the no of n's of the desired type is
$360 + 180 + 180 = 720$.

---

(3)(a) Define the well Ordering Principle. By Mathematical Induction, Prove that $(n!) \geq 2^{n-1}$ for all integers $n \geq 1$.

→ "Every non-empty subset of Z^+ contains a smallest (least) element" (OR)

"The Set of all positive integers is well Ordered".

Proof :-

Basic Step : For $n = 1$, $S(n)$ reads $(1!) \geq 2^{1-1}$ which is Obviously true. Thus $S(n)$ is verified for $n = 1$.

Induction Step :- We assume that $S(k)$ is true for $n = k$, where k is an integer ≥ 1; i.e, we assume that $(k!) \geq 2^{k-1}$ (OR) $2^{k-1} \leq k!$ is true

$$2^k = 2 \cdot 2^{k-1} \leq 2 \cdot k!$$
$$\leq (k+1) \cdot k! \text{ because } 2 < (k+1) \text{ for } k \geq 1$$
$$= (k+1)!$$
$$(k+1)! \geq 2^k$$

This is precisely the Statement $S(n)$ for $n = k+1$. Thus, On the assumption that $S(n)$ is true for $n = k \geq 1$, we have proved that $S(n)$ is true for $n = k+1$.

Hence, by MI, it follows the Statement $S(n)$ is true for all integers $n \geq 1$.

(b) Prove that every positive integer $n \geq 24$ can be written as a Sum of 5's and/(OR) 7's.

→ Basic Step :- We note that $24 = (7+7)+(5+5)$

This shows $S(24)$ is true.

Induction Step: We assume that $S(n)$ is true for $n=k$ where $k \geq 24$. Then

$k = (7+7+\dots) + (5+5+\dots)$

Suppose this representation of k has a no of 7's and s no of 5's. Since $k \geq 24$ we should have $a \geq 2$ and $s \geq 2$.

Using this representation of k, we find that

$$k+1 = \left\{ \underbrace{(7+7+\dots)}_{a} + \underbrace{(5+5+\dots)}_{s} \right\} + 1$$

$$= \left\{ \underbrace{(7+7+\dots)}_{a-2} + (7+7) + \underbrace{(5+5+\dots)}_{s} \right\} + 1$$

$$= \left\{ \underbrace{(7+7+\dots)}_{a-2} + \underbrace{(5+5+\dots)}_{s+3} \right\}$$

This shows that $k+1$ is sum of 7's and 5's.

Thus $S(k+1)$ is true.

(repeated)

(c) How many positive integers, n, can we form using the digits 3, 4, 4, 5, 5, 6, 7, if we want n to exceed 5,000,000

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(4) (a) By MI. P.T $1\cdot3 + 2\cdot4 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$

→ We have P.T $S(n) = 1\cdot3 + 2\cdot4 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$

Basic Step: We note that $S(1)$ is the Statement

$1\cdot3 = \frac{1}{6}\cdot1\cdot(1+1)\cdot(2+7)$

which is clearly true, the Statement $S(n)$ is verified for $n=1$.

Induction Step: We assume that the Statement $S(n)$ is true for $n=k$ where k is an integer ≥ 1; we assume Statement is true:

$S(k) = 1\cdot3 + 2\cdot4 + 3\cdot5 + \dots + k(k+2) = \frac{1}{6}k(k+1)(2k+7)$

Using this we find (by adding $(k+1)(k+3)$ to both sides)

$S(k) = 1\cdot3 + 2\cdot4 + \dots + k(k+2) + (k+1)(k+3) = \frac{1}{6}k(k+1)(2k+7) + (k+1)(k+3)$

$= (k+1)\left\{\frac{1}{6}k(2k+7)+(k+3)\right\}$

$= (k+1)\frac{1}{6}\{2k^2+7k+6k+18\}$

$= (k+1)\frac{1}{6}\{2k^2+13k+18\}$

$= \frac{1}{6}(k+1)(k+2)(2k+9)$

This is precisely the statement S(k+1). Thus, On the basis of assumption that S(n) is true, $n-k=1$, the truthness of S(n) for $n=k+1$ is established.

(b) Find the number of permutations of the letters of the word MASSASAUGA. In how many of these all four A's are together? How many of them begin with S?

→ The given word has 10 letters of which 4 are A, 3 are S and 1 each are M, U, G. The required no of permutations is $= \frac{10!}{4!3!1!1!1!} = 25200$

It is a permutation all A's are to be together, we treat all A's as one single letter. Then the letters to be permuted are (AAAA), S, S, S, M, U, G (which are 7 in no). The no of permutations is $= \frac{7!}{1!3!1!1!1!} = 840$

For permutations beginning with S, there occurs nine open positions to fill, where two are S, four are A & 1 each of M, U, G. The no of such permutations is

$= \frac{9!}{2!4!1!1!1!} = 7560$

(c) (i) Obtain the coefficient of a^5b^2 in expansion of $(2a-3b)^7$.

→ The binomial theorem states that:

$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$, $x=2a$, $y=-3b$, $n=7$.

exponent of $a=5$, $b=2$. [Substitute in BTJ.

$= (2a)^{7-k}(-3b)^k$ to contain a^5 & b^2 we need

$7-k=5$ and $\boxed{k=2}$.

Substitute $r=2$ into binomial expansion term:

$= \binom{7}{2}(2a)^{7-2}(-3b)^2$ | $\binom{7}{2} = \frac{7!}{2!(7-2)!} = \frac{7\times6}{2\times1} = 21$

$= 21\,(2a)^5(-3b)^2$

$\rightarrow (2a)^5 = 2^5a^5 = 32a^5$ [substitute back]

$\rightarrow (-3b)^2 = (-3)^2b^2 = 9b^2$

Combine these:

$= 21 \times 32a^5 \times 9b^2 = 21 \times 32 \times 9\, a^5b^2 = 6048\, a^5b^2$

∴ The coefficient of a^5b^2 in expansion of $(2a-3b)^7$ is 6048.

(ii) Using the Binomial theorem find the coefficient of x^5y^2 in expansion of $(x+y)^7$.

$\rightarrow (x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k}y^k$, $x=x$, $y=y$, $n=7$

$x=5$, $y=2$

$7-k=5 \rightarrow \boxed{k=2}$

Substitute x, y, n and $k=2$ values in $(x+y)^n$

$= \binom{7}{2} x^{(7-2)}\, y^2$ | $\binom{7}{2} = \frac{7\times6}{2\times1} = 21$

$= 21\,x^5y^2$

∴ The coefficient of x^5y^2 in expansion of $(x+y)^7$ is 21.

:- Module-3 :-

(5)(a) Let $f: R \to R$ be defined by, $f(x) = \begin{cases} 3x-5 & \text{if } x>0 \\ 1-3x & \text{if } x\le 0 \end{cases}$.

Find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(6)$, $f^{-1}([-6,5])$ and $f^{-1}([-5,5])$

$\rightarrow$ (i) $f^{-1}(0)$

$\rightarrow f^{-1}(0) = x$

$f(x) = 0$

$3x-5=0$

$x = 5/3 > 0$

$1-3x=0 \quad x = 1/3 > 0$

∴ $f^{-1}(0) = \{5/3\}$

(∵ $1-3x$, for $x\le 0$, but $x = 1/3 > 0$, which is not possible).

(iii) $f^{-1}(1)$

$\rightarrow f(x) = 1$

$3x - 5 = 1$

$3x = 6$

$x = 2 > 0$

$3 - 3x = 1$

$x = 0 \quad x \leq 0.$

$\therefore f^{-1}(1) = \{2, 0\}$

(iii) $f^{-1}(-1)$

$\rightarrow f(x) = -1$

$3x - 5 = -1$

$x = 4/3 > 0$

$1 - 3x = -1$

$x = 2/3 > 0$

$\therefore f^{-1}(-1) = \{4/3\}$

(iv) $f^{-1}(3)$

$\rightarrow f(x) = 3$

$3x - 5 = 3$

$x = 8/3 > 0$

$1 - 3x = 3$

$x = -2/3 < 0$

$\therefore f^{-1}(3) = \{8/3, -2/3\}$

(v) $f^{-1}(6)$

$\rightarrow f(x) = 6$

$3x - 5 = 6$

$x = 11/3 > 0$

$1 - 3x = 6$

$x = -5/3 > 0$

$f^{-1}(6) = \{11/3\}$

(vi) $f^{-1}([-6, 5])$

$\rightarrow f(x) = [-6, 5]$

$-6 \leq f(x) \leq 5$

$-6 \leq 3x - 5 \leq 5$

$-1 \leq 3x \leq 10$

$-1/3 \leq x \leq 10/3$

$x \in [-1/3, 10/3]$

$-6 \leq -3x + 1 \leq 5$

$-4 \leq -3x \leq 4$

$7/3 \geq x \geq -4/3$

$x \in [-4/3, 7/3]$

$x \in [-4/3, 10/3]$

(vii) $f^{-1}([-5, 5])$

$\rightarrow f(x) = [-5, 5]$

$-5 \leq f(x) \leq 5$

$-5 \leq 3x - 5 \leq 5$

$0 \leq 3x \leq 10$

$0 \leq x \leq 10/3$

$x \in [0, 10/3]$

$-5 \leq f(x) \leq 5$

$-5 \leq 1 - 3x \leq 5$

$-6 \leq -3x \leq 4$

$2 \geq x \geq -4/3$

$x \in [-4/3, 2]$

$x \in [-4/3, 10/3]$

(b) State Pigeon hole principle. P.T in any set of 99 persons at least 5 persons have been born on same day of the week.

→ If m pigeons occupy n pigeon holes and if $m > n$, then two (or) more pigeons occupy the same pigeon hole.

Generalizations: If m pigeons occupy n pigeon holes, then at least one pigeonhole must contain $(p+1)$ (or) more pigeons where $p = \left[\frac{m-1}{n}\right]$

The items are 89 persons.
The Containers are the 7 days of the week.
We want to P.T at least one of days of week must have at least 5 persons born on that day.
Since no of persons must be whole numbers, Some days will have 4 persons & Some will have 5. TO avoid any day having 5 persons, let's assume the most even distribution where each day has minimum no of persons possible.

If each day has 4 persons, then we have 89 persons & One extra person.

$28 + 1 = 29$

at least One day must have : $4 + 1 = 5$ persons.

∴ That in any Set of 89 persons, at least 5 persons must been born on Same day of week.

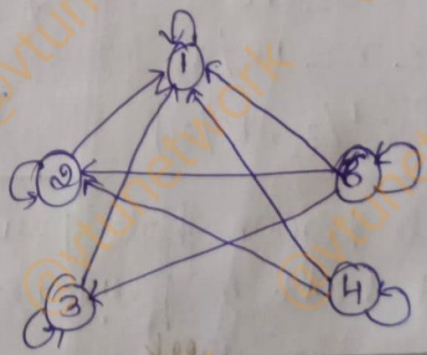
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(5) Let $A = \{1, 2, 3, 4, 6\}$ and 'R' be a relation on 'A' defined by aRb if & only if "a is multiple of b" represent the relation 'R' as a matrix, draw diagraph & relation R.

→ $R = \{(1,1), (2,1), (2,2), (3,1), (3,3), (4,1), (4,2), (4,4), (6,1), (6,2), (6,3), (6,6)\}$

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

→ Diagraph :-



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(6) (a) If $f: A \to B$, $g: B \to C$, $h: C \to D$ are three functions, then Prove that $ho(gof) = (hog)of$.

→ Since $f: A \to B$, $g: B \to C$ and $h: C \to D$, we have both $(hog)of$ and $ho(gof)$ are functions from A to D for any $x \in A$.

$[(ho(qof)](x) = h[(qof)(x)]$

$= h[g\{f(x)\}]$ $[y = f(x)]$

$= h[g(y)]$

$= h(z) \rightarrow ①$ $[z = g(y)]$

$[(hog)of](x) = (hog)[f(x)]$

$= (hog)(y)$

$= h[g(y)]$

$= h(z) \rightarrow ②$

∴ Thus $[(ho(qof)](x) = [(hog)of](x)$ for every $x \in A$,

∴ $ho(qof) = (hog)of$

∴ Compositions of three functions is associative.

(b) Show that if $n+1$ numbers are chosen from 1 to $2n$, then at least one pair add to $2n+1$.

→ Let we Consider the following Sets:

$A_1 = \{1, 2n\}$ $A_2 = \{2, 2n-1\}$, $A_3 = \{3, 2n-2\}$

$A_{n-1} = \{n-1, n+2\}$ $A_n = \{n, n+1\}$

There are the only Sets Containing 2 nos & from 1 to n whose Sum is $2n+1$.

Since there are only n-Sets, two of the $n+1$ choosen nos belong to some Set (by pigeon hole principle) & Sum of these 2 numbers = $2n+1$.

(c) Draw the Hasse diagram representing the positive divisors of 72.

→ The Set of positive divisors of 72 is

$D_{72} = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72\}$

The relation R of divisibility (that is aRb if & only if a divides b) is a partial Order on this Set! The Hasse diagram for partial Order is required here.

1 is related to all elements of D_{72}

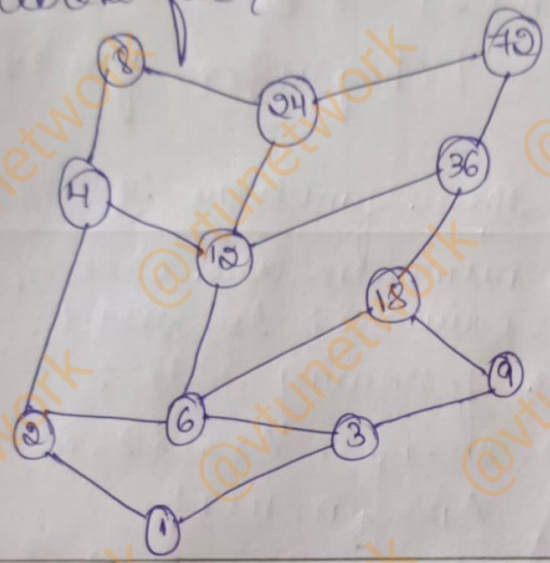
2 is related to 4, 6, 8, 12, 18, 24, 36, 72

3 is related to 9, 12, 6, 18, 24, 36, 72.

4 is related to 8, 12, 24, 36, 72

6 is related to 12, 18, 24, 36, 72
8 is related to 24, 72.
9 is related to 18, 36, 72
12 is related to 24, 36, 72
18 is related to 36, 72
24 is related to 72.
36 is related to 72

The Hasse diagram for R must exhibit all of the above facts.



(5) (a) State pigeon-hole principle. P.T if any nos from 1 to 8 are chosen then two of them will have their sum as 9

→ The pigeon hole principle states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item.

The pairs of numbers that sum to 9 within range 1 to 8 are: {(1,8), (2,7), (3,6), (4,5)}

If we select any 5 nos from the set {1, 2, 3, 4, 5, 6, 7, 8} we have more nos than pairs.

According to pigeon hole principle, if we pick 5 nos from these 8, at least one of these pairs must be chosen bcz we have only 4 pairs & we choosing 5 nos. Hence guarantee that at least two of chosen nos will sum to 9.

∴ It's impossible to select 5 nos from set {1, 2, 3, 4, 5, 6, 7, 8} without having at least one pair that

sums to 9. Hence, if any 5 numbers are chosen from set, at least two of them will have sum as 9.

(b) Let $f: R \to R$ be defined by $f(x) = \begin{cases} 3x-5, & \text{if } x > 0 \\ 1-3x, & \text{if } x \le 0 \end{cases}$ find $f^{-1}([-6,5])$ and $f^{-1}([-5,5])$

(c) Let $A = B = C = R$, and $f: A \to B$ and $g: B \to C$ be defined by $f(a) = 2a+1$, $g(b) = \frac{1}{3}b$, $\forall a \in A, b \in B$. Compute gof & show that gof is invertible. what is $(gof)^{-1}$?

→ We have $(gof)(a) = g\{f(a)\} = g(2a+1) = \frac{1}{3}(2a+1)$

$gof: A \to C$ is defined by $(gof)(a) = \frac{1}{3}(2a+1)$

we have $b = f(a) = 2a+1 \Rightarrow a = \frac{b-1}{2}$

$\therefore a = f^{-1}(b) = \frac{b-1}{2} \Rightarrow f$ is ~~invertible~~ invertible.

Also $c = g(b) = \frac{1}{3}b \Rightarrow b = 3c$

$\therefore b = g^{-1}(c) = 3c \Rightarrow g$ is invertible

Thus gof is invertible and its inverse is given by

$(gof)^{-1}(c) = (f^{-1}og^{-1})(c) = f^{-1}\{g^{-1}(c)\} = f^{-1}(3c) = \frac{3c-1}{2}$

(6) (a) Let f and g be functions from R to R defined by $f(x) = ax+b$ and $g(x) = 1-x+x^2$. If $(gof)(x) = 9x^2 - 9x + 3$. determine a and b.

→ Given $f(x) = ax+b$, $g(x) = 1-x+x^2$

$(gof)x = g(f(x)) = g(ax+b) = 1-ax+b+(ax+b)^2$

$= 1-ax+b+a^2x^2+b^2+2abx \Rightarrow a^2x^2 + x(2ab-a) + (1-b+b^2)$

$(gof)x = a^2x^2 + x(2ab-a) + (1-b+b^2)$

W.K.T by data

$gof(x) = 9x^2 - 9x + 3 = a^2x^2 + x(2ab-a) + (1-b+b^2)$

On equating the like terms

$a^2 = 9$ $\quad 2ab - a = -9$

$\boxed{a = \pm 3}$

If $a = 3$

$6b - 3 = -9$

$6b = -6$

$\boxed{b = -1}$

If $a = -3$

$-6b + 3 = -9$

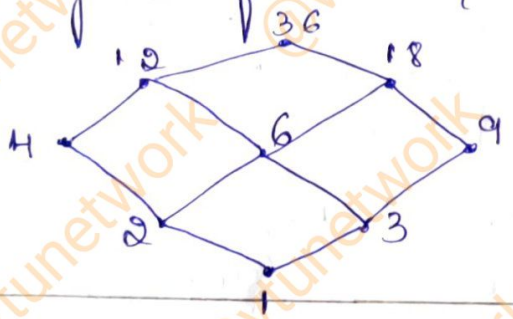
$\boxed{b = 2}$

(b) Draw the Hasse (poset) diagram which represents positive divisors of 36.

→ The set of positive divisors of 36 is $D_{36} = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$. The relation R of divisibility is partial order on this set. The Hasse diagram for partial order is required here.

1 is related to all elements of D_{36}
2 is related to 2, 4, 6, 12, 18, 36
3 is related to 3, 6, 9, 12, 18, 36
4 is related to 4, 12, 36
6 is related to 6, 12, 18, 36
9 is related to 9, 18, 36
12 is related to 12 & 36
18 is related to 18 & 36
36 is related to 36

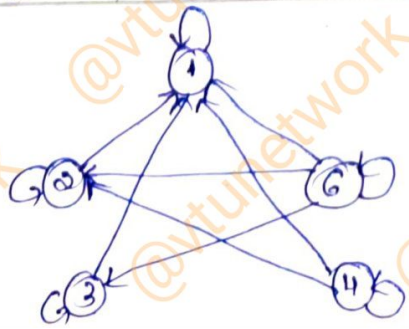
The Hasse diagram for R must exhibit all of above facts:



(c) Let $A = \{1, 2, 3, 4, 6\}$ & R be relation on A defined by aRb if & if "a is a multiple of b". Write down the relation R, relation matrix M(R) and draw its diagraph. List out its in degree and out degree.

→

Vertex	1	2	3	4	6
In-degree	5	3	2	1	1
Out-degree	1	2	2	3	4



-: MOdule-4 :-

(7)(a) In how many ways the 26 letters of English alphabet are permuted so that none of the patterns CAR, DOG, PUN (or) BYTE occurs?

→ Let S denote the set of all permutations of the 26 letters. Then $|S| = 26!$

Let A_1 be the set of all permutations in which CAR appears. This would CAR consists of 3 letters which form a single block.

Similarly, if A_2, A_3, A_4 are set of all permutations which contain DOG, PUN & BYTE respectively.

$|A_2| = 24!$ $|A_3| = 24!$ $|A_4| = 23!$

$|A_1 \cap A_2| = |A_1 \cap A_3| = |A_2 \cap A_3| = (26-6+2)! = 22!$

$|A_1 \cap A_4| = |A_2 \cap A_4| = |A_3 \cap A_4| = (26-7+2)! = 21!$

$|A_1 \cap A_2 \cap A_3| = (26-9+3)! = 20!$

$|A_1 \cap A_2 \cap A_4| = |A_1 \cap A_3 \cap A_4| = |A_2 \cap A_3 \cap A_4| = (26-10+3)! = 19!$

$|A_1 \cap A_2 \cap A_3 \cap A_4| = (26-13+4)! = 17!$

∴ the required no of permutations is given by,

$|\bar{A_1} \cap \bar{A_2} \cap \bar{A_3} \cap \bar{A_4}| = |S| - \Sigma |A_i| + \Sigma |A_i \cap A_j| - \Sigma |A_i \cap A_j \cap A_k| + |A_1 \cap A_2 \cap A_3 \cap A_4|$

$= 26! - (3 \times 24! + 23!) + (3 \times 22! + 3 \times 21!) - (20! + 3 \times 19!) + 17!$

(b) Define Derangement. In how many ways each of 10 people select a left glove & a right glove out of a total of 10 pairs of gloves so that no person selects a matching pair of gloves.

→ A permutation of n distinct Objects in which none of the Objects is in its natural place is called derangement.

10 people can select a left glove & right glove such that no person selects a matching pair of gloves.

① Choosing left gloves → The first person has 10 choices for their left glove. The second person can choose

from remaining 9 pairs, The no of ways to choose left gloves for all 10 people without any restriction is $10! = 10\times9\times8\times7\times6\times5\times4\times3\times2\times1 = 3628800$

(b) Choosing right gloves:
Identifying total possible Selections: Each person has 10 choices for left, 10 choices for right glove.
$10\times10 = 100$ possible pairs per person.
Removing matching pairs:- Since no one can select a matching pairs, Subtract 10 matching pairs.
Total Selections without matching pairs :- $100-10=90$ possible pairs per person.
The no of derangements in of n items given by:

$$!n = n!\sum_{i=0}^{n}\frac{(-1)^i}{i!} \rightarrow n=10 \rightarrow 10! = 10!\left(\sum_{i=0}^{10}\frac{(-1)^i}{i!}\right)$$

$10! = 3628800$

Sum of series: $\sum_{i=0}^{10}\frac{(-1)^i}{i!} = 1-1+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}-\frac{1}{7!}+\frac{1}{8!}-\frac{1}{9!}+\frac{1}{10!}$

$\approx 1-1+0.5-0.1667+0.0417-0.0083+0.0014-0.0002+0.0002-0.000002+0.000002 \simeq 0.3679$

$!10 = 3628800\times0.3679 \simeq 1,334,961$

∴ The no of ways 10 people can select a left glove & a right glove. Out of 10 pairs of gloves is approximately 1,334,961

(c) Solve the recurrence relation: $C_n = 3C_{n-1} - 2C_{n-2}$, for $n\geq2$, given $C_1=5$, $C_2=3$.

→ Assume $C_n = u^n = 3u^{n-1} - 2u^{n-2}$ ÷ by $[u^{n-2}$ (for $u\neq0)]$

$u^2 = 3u-2 \rightarrow u^2-3u+2=0$

$u = \frac{3\pm\sqrt{9-8}}{2} = \frac{3\pm1}{2} \rightarrow$ $\boxed{u_1=2}$ $\boxed{u_2=1}$

General Solution of recurrence relation is:
$C_n = A\cdot2^n + B\cdot1^n$

Since $1^n = 1 \rightarrow C_n = A \cdot 2^n + B$

for $C_1 = 5 \Rightarrow 5 = A \cdot 2^1 + B \Rightarrow 5 = 2A + B \rightarrow (1)$

for $C_2 = 3 \Rightarrow 3 = A \cdot 2^2 + B \Rightarrow 3 = 4A + B \rightarrow (2)$

Subtract eqn (1) from eqn (2)

$(4A + B) - (2A + B) = 3 - 5 \Rightarrow 2A = -2 \quad \boxed{A = -1}$

Substitute $A = -1$ in eqn (1)

$5 = 2(-1) + B \rightarrow -2 + B = 5 \rightarrow \boxed{B = 7}$

Substitute A & B in General equation.

$C_n = -1 \cdot 2^n + 7 \rightarrow C_n = 7 - 2^n$

∴ The solution to recurrence relation $C_n = 3C_{n-1} - 2C_{n-2}$ with given initial conditions $C_1 = 5$ & $C_2 = 3$ is $C_n = 7 - 2^n$.

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(8) (a) In how many ways one can arrange the letters of the word CORRESPONDENTS so that there are (i) exactly 2 pairs of consecutive identical letters (ii) at least 3 pairs of consecutive identical letters. (iii) no pair of consecutive identical letters?

→ In the word CORRESPONDENTS, there occur one each of C, P, D & T & two each of O, R, E, S, N. If S is the set of all permutations of these 14 letters. $|S| = \frac{14!}{(2!)^5}$

Let A_1, A_2, A_3, A_4, A_5 be set of all permutations in which O's, R's, E's, N's appear in pairs respectively.

$|A_i| = \frac{13!}{(2!)^4}$ for $i = 1, 2, 3, 4, 5$

$|A_i \cap A_j| = \frac{12!}{(2!)^3}$, $|A_i \cap A_j \cap A_k| = \frac{11!}{(2!)^2}$, $|A_i \cap A_j \cap A_k \cap A_p| = \frac{10!}{(2!)}$

$|A_1 \cap A_2 \cap A_3 \ldots \cap A_5| = 9!$

$S_0 = N = |S| = \frac{14!}{(2!)^5}$, $S_1 = C(5,1) \times \frac{13!}{(2!)^4}$

$S_2 = C(5,2) \times \frac{12!}{(2!)^3}$, $S_3 = C(5,3) \times \frac{11!}{(2!)^2}$

$S_4 = C(5,4) \times \frac{10!}{(2!)^1}$, $S_5 = C(5,5) \times 9!$

(i) The no of permutations where exactly two pairs of consecutive identical letters.

$$E_2 = S_2 - \binom{3}{1}S_3 + \binom{4}{2}S_4 - \binom{5}{3}S_5$$

$$= \binom{5}{2} \times \frac{12!}{(2!)^2} - \binom{3}{1}\binom{5}{3} \times \frac{11!}{(2!)^3} + \binom{4}{2}\binom{5}{4} \times \frac{10!}{(2!)^1} - \binom{5}{3}\binom{5}{5} \times 9!$$

(ii) The no of permutations where at least three pairs of consecutive identical letters

$$E_3 = S_3 - \binom{3}{2}S_4 + \binom{4}{3}S_5$$

$$= \binom{5}{3} \times \frac{11!}{(2!)^2} + \binom{3}{2}\binom{5}{4} \times \frac{10!}{(2!)^1} - \binom{4}{2}\binom{5}{5} \times 9!$$

(iii) The no of permutations where there is no pair of consecutive identical letters

$$E_0 = S_0 - \binom{1}{1}S_1 + \binom{2}{2}S_2 - \binom{3}{3}S_3 + \binom{4}{4}S_4 - \binom{5}{5}S_5$$

$$= \frac{14!}{(2!)^5} - \binom{5}{1} \times \frac{13!}{(2!)^4} + \binom{5}{2} \times \frac{12!}{(2!)^3} - \binom{5}{3} \times \frac{11!}{(2!)^2} + \binom{5}{4} \times \frac{10!}{(2!)^1} - \binom{5}{5} \times 9!$$

(b) Find the rook polynomial for the chess board as shown in the fig



→ We need to determine the no of ways to place k non-attacking rooks on board for each k.
The fig appears to be an L-shaped with four squares

(i) 0 rooks: There's exactly 1 way to do this.

(ii) 1 rooks: We can place 1 rook in any 4 squares. This gives 4 ways.

(iii) 2 rooks: We need to place 2 rooks on 2 non-attacking squares
- The rook in top-left can pair with any 3 squares.
- The rook in top-middle can pair with bottom left / right squares.
- The rook in bottom-left can pair with top-middle square

This sums up to 3 unique configuration.

(iv) 3 rooks: Placing 3 non-attacking rooks each rook must

be in separate row & column. There are 4 ways.

(iv) 4 rooks : This configuration is not possible, we cannot place 4 non-attacking rooks on this board.

$a_0 = 1$, $a_1 = 4$, $a_2 = 3$, $a_3 = 0$, $a_4 = 0$

So rook polynomial $R(x)$ for board is:

$R(x) = 1 + 4x + 3x^2$

(c) Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0$, $a_0 = 1$, $a_1 = 6$.

→ Characteristic eqn for recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0$ is: $r^2 - 3r + 2 = 0$

$r^2 - 3r + 2 \Rightarrow (r-1)(r-2) = 0$

$\therefore \boxed{r_1 = 1}$ $\boxed{r_2 = 2}$

$r^2 - 3r + 2 = 0$
$r^2 - 1r - 2r + 2 = 0$
$r(r-1) - 2(r-1) = 0$
$(r-2)(r-1) = 0$

General Soln to recurrence relation is:

$a_n = A \cdot 1^n + B \cdot 2^n = A + B \cdot 2^n$

$a_0 = 1 \Rightarrow A + B \cdot 2^0 = 1 \Rightarrow A + B = 1 \rightarrow (1)$

$a_1 = 6 \Rightarrow A + B \cdot 2^1 = 6 \Rightarrow A + 2B = 6 \rightarrow (2)$

Subtract eqn (1) from eqn (2)

$(A + 2B) - (A + B) = 6 - 1$

$A + 2B - A - B = 5 \Rightarrow \boxed{B = 5}$

Substitute $B = 5$ into eqn (1) $\Rightarrow A + 5 = 1$ $\boxed{A = -4}$

$\boxed{a_n = -4 + 5 \cdot 2^n}$ [Substitute A & B values in General Soln]

∴ The Soln to recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0$ with initial conditions is: $a_n = -4 + 5 \cdot 2^n$

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(7)(a) Determine the no of +ve integers n such that $1 \le n \le 100$ & n is not divisible by 2, 3 (or) 5.

→ $S = \{1, 2, 3 \ldots 100\}$ ∴ $|S| = 100$ ∴ $S_0 = 100$

Let A = Set of nos divisible by 2, $B = 3$, $C = 5$

$|\bar{A} \cap \bar{B} \cap \bar{C}|$ is $|\overline{A \cup B \cup C}|$

∴ $|A|$ = no of elements in S that are divisible by 2.

$|A| = \lfloor \frac{100}{2} \rfloor = 50$

$|B| = \lfloor \frac{100}{3} \rfloor = \lfloor 33.33 \rfloor = 33$

$|C| = \lfloor \frac{100}{5} \rfloor = 20$

$\therefore S_1 = |A| + |B| + |C| = 50 + 33 + 20 = 103$

$|A \cap B|$ = no of elements in S divisible by both 2 & 3.

$|A \cap B| = \left\lfloor \frac{100}{6} \right\rfloor = \lfloor 16.66 \rfloor = 16$

$|A \cap C| = \left\lfloor \frac{100}{10} \right\rfloor = 10 \quad |B \cap C| = \left\lfloor \frac{100}{15} \right\rfloor = 6$

$\therefore S_2 = 16 + 10 + 6 = 32$

$|A \cap B \cap C|$ = no of elements in S divisible by all 2, 3, 5

$= \left\lfloor \frac{100}{30} \right\rfloor = 3 \quad \therefore S_3 = 3$

$\therefore |\bar{A} \cap \bar{B} \cap \bar{C}| = |S| - (|A| + |B| + |C|) + (|A \cap B| + |B \cap C| + |A \cap C|) - |A \cap B \cap C|$

$= 100 - 103 + 16 + 10 + 6 - 3 = 26.$

(repeated)

(b) In how many ways can the 26 letters of English alphabet be permuted so that none of patterns CAR, DOG, PUN or BYTE occurs.

(c) Solve the recurrence relation $a_n = n a_{n-1}$ where $n \geq 1$ & $a_0 = 1$.

→ $a_1 = 1 \times a_0$, $a_2 = 2a_1 = (2 \times 1)a_0$

$a_3 = 3 \times a_2 = (3 \times 2 \times 1)a_0$, $a_4 = 4 \times a_3 = (4 \times 3 \times 2 \times 1)a_0$.

The General solution is $a_n = (n!)a_0$ for $n \geq 1$

Using initial Condition $a_0 = 1$.

$\therefore a_n = n!$

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(8) (a) In how many ways 5 no of a's, 4 no of b's & 3 no of c's can be arranged so that all the identical letters are not in a single block?

→ The given letters are $5 + 4 + 3 = 12$ of which 5 are a's, 4 are b's, 3 are c's.

$|S| = \frac{12!}{5!4!3!}$.

Let A_1 be the set of arrangements of letters where the 5 a's are in single block, $|A_1| = \frac{8!}{4!3!}$

A_2 be the set of arrangements of letters where the 4 b's are in single block, $|A_2| = \frac{9!}{5!3!}$

A_3 is the set of arrangements of letters where 3 C's are in single block $|A_3| = \frac{10!}{5!4!}$

$|A_1 \cap A_2| = \frac{5!}{3!}$, $|A_1 \cap A_3| = \frac{6!}{4!}$, $|A_2 \cap A_3| = \frac{7!}{5!}$

$|A_1 \cap A_2 \cap A_3| = 3!$

The required no of arrangements is, $|\bar{A_1} \cap \bar{A_2} \cap \bar{A_3}| =$

$= |S| - \{|A_1 \cup A_2 \cup A_3|\} + \{|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|\} - |A_1 \cap A_2 \cap A_3|$

$= \frac{12!}{5!4!3!} - \left\{\frac{8!}{4!3!} + \frac{9!}{5!3!} + \frac{10!}{5!4!}\right\} + \left\{\frac{5!}{3!} + \frac{6!}{4!} + \frac{7!}{5!}\right\}$

$= 27720 - (280 + 504 + 1260) + (20 + 30 + 42) - 6 = 25762$

(b) Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers are five classes, C_1, C_2, C_3, C_4, C_5, One teacher for One class. T_1 & T_2 do not wish to become the class teacher for C_1 (or) C_2, T_3 & T_4 for C_4 (or) C_5, & T_5 for C_3 (or) C_4 (or) C_5. In how many ways can teachers be assigned the work (without displeasing any teacher).

→ $5! = 120$ [Total permutations without any restriction]

Let A be set of permutations where T_1 is assigned to C_1 (or) C_2. B be set of permutations where T_2 is assigned to C_1 (or) C_2.

C be permutations where T_3 is assigned C_4 (or) C_5

D be permutations where T_4 is assigned C_4 (or) C_5

E be permutations where T_5 is assigned to C_3 (or) C_4, C_5

$|A \cup B \cup C \cup D \cup E| = \Sigma |A_i| - \Sigma |A_i \cap A_j| + \Sigma |A_i \cap A_j \cap A_k| - \Sigma |A_i \cap A_j \cap A_k \cap A_l| + |A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5|$

$|A| = |B| = 2 \cdot 4! = 48$, $|C| = |D| = 2 \cdot 4! = 48$, $|E| = 3 \cdot 4! = 72$

$|A \cap B| = 2 \cdot 2 \cdot 3! = 24$, $|A \cap C| = |A \cap D| = |B \cap C| = |B \cap D| = 2 \cdot 2 \cdot 3! = 24$

$|A \cap E| = |B \cap E| = 2 \cdot 3 \cdot 3! = 36$, $|C \cap E| = |D \cap E| = 2 \cdot 3 \cdot 3! = 36$

$|C \cap D| = 2 \cdot 2 \cdot 3! = 24$, $|A \cap B \cap C| = |A \cap B \cap D| = 2 \cdot 2 \cdot 2 \cdot 2! = 8$

$|A \cap B \cap E| = 2 \cdot 2 \cdot 3 \cdot 2! = 24$, $|C \cap D \cap E| = 2 \cdot 2 \cdot 3 \cdot 2! = 24$

$|A \cup B \cup C \cup D \cup E| = 240 - 240 + 96 - 24 = 72$.

Valid permutations = 100 - 72 = 48
There are 48 ways to assign the teachers to classes without disposing any teacher.

(8) Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ where $n \geq 0$ & $F_0 = 0$, $F_1 = 1$

$\rightarrow f_n = F_{n-1} + F_{n-2}$

$f_n - F_{n-1} - f_{n-2} = 0 \rightarrow$ (1) for $n \geq 2$

$k^2 - k - 1 = 0 \rightarrow k = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$

General soln of eqn (1) is

$f_n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n$ ... (2) where A & B are arbitrary constants

$F_0 = 0, F_1 = 1$

Sub $n=0$ in (2) $F_0 = A + B \Rightarrow \boxed{0 = A + B}$ → (3)

Substitute $n=1$ in (2), $F_1 = A\left(\frac{1+\sqrt{5}}{2}\right) + B\left(\frac{1-\sqrt{5}}{2}\right)$

$\Rightarrow \boxed{1 = A\left(\frac{1+\sqrt{5}}{2}\right) + B\left(\frac{1-\sqrt{5}}{2}\right)}$ → (4)

Eqn (3) $\times \left(\frac{1+\sqrt{5}}{2}\right)$ gives $\rightarrow 0 = A\left(\frac{1+\sqrt{5}}{2}\right) + B\left(\frac{1+\sqrt{5}}{2}\right)$ → (5)

(4) - (5) $\rightarrow 1 = B\left(\frac{1-\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}\right) = B\left(\frac{-2\sqrt{5}}{2}\right) \rightarrow \boxed{B = -1/\sqrt{5}}$

from (3) $A = -B \Rightarrow \boxed{A = 1/\sqrt{5}}$

Substitute in (2),

$$\boxed{F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]}$$

: Module-5 :-

(9) (a) If H, K are subgroups of a group G, P.T $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G.

→ (i) closure: For any $h, k \in H \cap K$, $h, k \in H$ & $h, k \in K$. Since H & K are subgroups, $h, k \in H$ & $hk \in K$

$\therefore hk \in H \cap K$, ensuring closure under group operation in G.

(2) Identity Element : H & K being subgroups imply they Conta-in the identity element e of G. $e \in H \cap K$.

(3) Inverse Element : If $h \in H \cap K$, then $h \in H$ & $h \in K$. H & K are subgroups, $h^{-1} \in H$ & $h^{-1} \in K$. $h^{-1} \in H \cap K$, ensuring existence of inverses.

Since $H \cap K$ satisfies all three Criteria, it is subgroup of G.

$H \cup K$ it is not necessarily a subgroup of G. The union of two subgroups H & K may fail the subgroup Criteria.

(1) Closure : There is no guarantee for $h, k \in H \cup K$, their product hk also be in $H \cup K$. $H \cup K$ might lack Closure under group Operation if hk falls Outside $H \cup K$.

(2) Identity element : $H \cup K$ may not Contain identity element e of G.

(3) Inverse element : $H \cup K$ may not Contain inverses for all elements within it.

∴ $H \cup K$ does not generally satisfy the subgroup Criteria & hence it not a subgroup of G.

(b) Define klein 4 group. Verify $A = \{1, 3, 5, 7\}$ is a klein 4 group.

→ The klein 4-group is a group with four elements in which every element is its own inverse & group operation is Commutative.

To verify the set $A = \{1, 3, 5, 7\}$ forms a klein 4-group under multiplication modulo 8, we need to check the following Criteria :

(1) The set has exactly 4 elements
(2) Each element has its own inverse.
(3) The group Operation is Commutative
(4) The product of any two distinct elements is the third element.

(1) Cardinality : $A = \{1, 3, 5, 7\}$ has exactly 4 elements.

(2) Inverses : The inverse of 1 is 1 bcz $1 \times 1 \equiv 1 \bmod 8$
The inverse of 3 is 3 bcz $3 \times 3 = 9 \equiv 1 \bmod 8$.
The inverse of 5 is 5 bcz $5 \times 5 \equiv 25 \equiv 1 \bmod 8$.

The inverse of 7 is 7 bcz $7 \times 7 = 49 \equiv 1 \bmod 8$.
∴ Each element is its own inverse.

③ Commutativity: $1 \times 3 \equiv 3 \times 1 \bmod 8$
$1 \times 5 \equiv 5 \times 1 \bmod 8$, $1 \times 7 \equiv 7 \times 1 \bmod 8$
$3 \times 5 \equiv 15 \equiv 7 \bmod 8$, $3 \times 7 \equiv 21 \equiv 5 \bmod 8$
$5 \times 7 \equiv 35 \equiv 3 \bmod 8$.
∴ multiplication is Commutative for elements in A.

④ Group Structure: $3 \times 5 \equiv 7 \bmod 8$,
$3 \times 7 \equiv 5 \bmod 8$, $5 \times 7 \equiv 3 \bmod 8$, $1 \times 3 \equiv 3 \bmod 8$,
$1 \times 5 \equiv 5 \bmod 8$, $1 \times 7 \equiv 7 \bmod 8$.
∴ Set $A = \{1, 3, 5, 7\}$ satisfies all properties of a klein 4-group.

© State and prove Lagrange's theorem.
→ Statement: Let G be finite group & H be subgroup of G. Then the Order of H divides the Order of G.
$|G| = [G:H] \cdot |H|$, where $[G:H]$ is index of H in G.

Proof:
① Cosets & Partitioning: A Coset of H in G is a set of the form $gH = \{gh \mid h \in H\}$ for $g \in G$.
The group G can be partitioned into disjoint left Cosets of H. G is exactly one left Coset of H.

② Size of Each Coset: Each left Coset gH has exactly $|H|$ elements bcz the map $h \mapsto gh$ is bijection from H to gH.

③ Counting elements: $|G| = [G:H] \times |H|$

④ Conclusion: $|G| = [G:H] \times |H|$, it follows $|H|$ divides $|G|$.

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10 (a) S.T (i) the identity of G is unique.
→ By defⁿ of identity element, for any $g \in G$:
$g \cdot e = g$ and $e' \cdot g = g$.
Consider the element e & use the property of e':
$e' \cdot e = e$ (e is an identity element)

Also, using the property of e: $e' \cdot e = e'$

$\therefore e = e'$

Since e & e' are equal, the identity element in G must be unique.

(ii) The inverse of each element in G is unique.

→ $g \in G$ has two inverses, h and h'

By defn of inverse, we have $g \cdot h = e$ & $h \cdot g = e$

$g \cdot h' = e$ & $h' \cdot g = e$

Consider $h \cdot (g \cdot h')$ using associativity

$h \cdot (g \cdot h') = (h \cdot g) \cdot h' = e \cdot h' = h'$

Using the fact $g \cdot h' = e$:

$h \cdot (g \cdot h') = h \cdot e = h$

$\therefore h = h'$

Since h & h' are equal, the inverse of any element $g \in G$ must be unique.

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(6) S.T $(A, \cdot)$ is an abelian group where $A = \{a \in \mathbb{Q} \mid a \neq -1\}$ & for any $a, b \in A$, $a \cdot b = a + b + ab$.

→ (1) Closure: S.T for any $a, b \in A$, $a \cdot b \in A$.

Let $a, b \in A$. then $a \cdot b = a + b + ab$

$a + b + ab \neq -1$

Assume $a \cdot b = -1$:

$a + b + ab = -1 \Rightarrow ab + a + b = -1 \Rightarrow ab + a + b + 1 = 0$

$\Rightarrow (a+1)(b+1) = 0$

Since $a \neq -1$ & $b \neq -1$, $(a+1) \neq 0$, $(b+1) \neq 0$.

$\therefore (a+1)(b+1) \neq 0$, $a \cdot b \neq -1$

$\therefore a \cdot b \in A$, So A is closed under operation.

(2) Associativity: ST $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in A$

$(a \cdot b) \cdot c = (a + b + ab) \cdot c = (a + b + ab) + c + (a + b + ab)c$

$= a + b + ab + c + ac + bc + abc$

$a \cdot (b \cdot c) = a \cdot (b + c + bc) = a + (b + c + bc) + a(b + c + bc)$

$= a + b + c + bc + ab + ac + abc$

$\therefore$ both expressions are equal. It is associative.

(3) Identity element: $e \in A$ such that $a \cdot e = a$ & $e \cdot a = a$ for all $a \in A$

$a \cdot e = a + e + ae = a$:

$a + e + ae = a \Rightarrow e + ae = 0 \Rightarrow e(1+a) = 0$

Since $a \neq -1$, $1+a \neq 0$ So $e = 0$. is the identity element

(4) Inverses: ST $a \in A$, exists an element $b \in A$ such that $a \cdot b = 0$ & $b \cdot a = 0$.

$a \cdot b = a + b + ab = 0$;

$a + b + ab = 0 \Rightarrow b(1+a) = -a \Rightarrow b = \frac{-a}{1+a}$

Since $a \neq -1$, $1+a \neq 0$, & $\frac{-a}{1+a} \in \mathbb{Q}$, $b \in A$.

(5) Commutativity: ST $a \cdot b = b \cdot a$ for all $a, b \in A$.

$a \cdot b = a + b + ab$

$b \cdot a = b + a + ba = a + b + ab$ ∴ Operation is Commutative.

$(A, .)$ is an abelian group.

---

(6) Let $G = S_4$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find subgroup of $H = \langle \alpha \rangle$. Determine left cosets of H in G.

→ $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$ $\alpha = (1\ 2\ 3\ 4)$

$H = \langle \alpha \rangle = \{ \alpha^k \mid k \in \mathbb{Z} \}$

$\alpha^0 = id = ()$, $\alpha^1 = (1\ 2\ 3\ 4)$, $\alpha^2 = (1\ 3)(2\ 4)$

$\alpha^3 = (1\ 4\ 3\ 2)$, $\alpha^4 = id = ()$

$H := \{ (), (1\ 2\ 3\ 4), (1\ 3)(2\ 4), (1\ 4\ 3\ 2) \}$

(1) $H = \{ (), (1\ 2\ 3\ 4), (1\ 3)(2\ 4), (1\ 4\ 3\ 2) \}$

$(1\ 2)H = \{ (1\ 2), (1\ 2)(1\ 2\ 3\ 4), (1\ 2)(1\ 3)(2\ 4), (1\ 2)(1\ 4\ 3\ 2) \}$

$(1\ 2)(1\ 2\ 3\ 4) = (2\ 3\ 4)$, $(1\ 2)(1\ 3)(2\ 4) = (1\ 4\ 2\ 3)$,

$(1\ 2)(1\ 4\ 3\ 2) = (1\ 3\ 4)$

$(1\ 2)H = \{ (1\ 2), (2\ 3\ 4), (1\ 4\ 2\ 3), (1\ 3\ 4) \}$

$(1\ 3)H = \{ (1\ 3), (1\ 3\ 2\ 4), (1\ 2\ 4\ 3), (1\ 4) \}$

$(14)H = \{(14), (1432), (1234), (13)\}$

$(12)(34)H = \{(12)(34), (24), (1423), (13)(24)\}$

$(13)(24)H = \{(13)(24), (24)(13), (12)(34), (14)\}$

There are left Cosets of H in S_4.

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(9)(a) Define group. S.T fourth roots of unity is an abelian group.

→ A group is a set G equipped with binary Operation that satisfies four fundamental properties.

The fourth roots of unity are Complex nos that satisfy eqn $z^4 = 1$. roots are given by: 1, i, -1, -i

(1) Closure: $\{1, i, -1, -i\}$

$1 \cdot 1 = 1$, $1 \cdot i = i$, $1 \cdot -1 = -1$, $1 \cdot -i = -i$

$i \cdot i = -1$, $i \cdot -1 = -i$, $i \cdot -i = 1$

$-1 \cdot -1 = 1$, $-1 \cdot -i = i$, $-i \cdot -i = -1$

All products are in Set.

(2) Associativity: Multiplication of Complex no is associative $a, b, c \in \{1, i, -1, -i\}$ i.e $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

(3) Identity element: The identity element for multiplication is 1. $a \in \{1, i, -1, -i\}$.

$1 \cdot a = a \cdot 1 = a$

(4) Inverse: $1^{-1} = 1$, $i^{-1} = -i$, $(-1)^{-1} = -1$, $(-i)^{-1} = i$

This is bcz

$1 \cdot 1 = 1$, $i \cdot -i = 1$, $-1 \cdot -1 = 1$, $-i \cdot i = 1$

(5) Commutative: $a \cdot b = b \cdot a$

Since the set of fourth roots of unity satisfies 4 properties.

(b) If G be a set of all non-zero real no & let $a * b = \frac{ab}{2}$. S.T $(G, *)$ is an abelian group.

→ (1) Closure: Since a & b are non-zero real nos, their product ab is also non-zero real no. Dividing by 2, a non-zero Constant, still results in non-zero

real no. $\therefore a * b = \frac{ab}{2} \in G$. G is closed under *.

② Associative:

$(a * b) * c = \left(\frac{ab}{2}\right) * c = \frac{\left(\frac{ab}{2}\right)}{2} c = \frac{abc}{4}$

$a * (b * c) = a * \left(\frac{bc}{2}\right) = \frac{a\left(\frac{bc}{2}\right)}{2} = \frac{abc}{4}$

$\therefore$ Operation * is associative.

③ Identity:

$a * e = \frac{ae}{2} \Rightarrow a \Rightarrow e = 2$

$a * 2 = \frac{a \cdot 2}{2} \Rightarrow a$ & $2 * a = \frac{2 \cdot a}{2} = a$

④ Inverse: $a * b = 2$:

$a * b = \frac{ab}{2} = 2 \Rightarrow ab = 4 \Rightarrow b = \frac{4}{a}$

$a \in G$, $\frac{4}{a} \in G$ is inverse of a

⑤ Commutative: $a * b = \frac{ab}{2}$ & $b * a = \frac{ba}{2}$

$\therefore a * b = b * a$.

$(G, *)$ is an abelian group.

---

⑥ Define klein 4-group. And if $A = \{e, a, b, c\}$ then Show that this is a klein-4 group.

→ ① Identity: There is an identity element e such that any $x \in A$, $e \cdot x = x \cdot e = x$.

② Inverses: Every element is its own inverse i.e $x \cdot x = e$ for all $x \in A$.

③ Commutative: The group Operation is Commutative i.e $x \cdot y = y \cdot x$ for all $x, y \in A$.

④ Order: The group has exactly four elements.

$\therefore A = \{e, a, b, c\}$ satisfies all properties of klein 4-group. it is indeed a klein 4-group.

6 (a) Define Cyclic group & S.T (G, *) whose multiplication table as given below is Cyclic

*	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	d	e	f	a
c	c	d	e	f	a	b
d	d	e	f	a	b	c
e	e	f	a	b	c	d
f	f	a	b	c	d	e

→ A Group G is Cyclic such that every element of G can be expressed as g^n for some integer n. $g \in G$

Start with element a:

① a ② $a^2 = a$ ③ $a^3 = a$ ④ $a^4 = a$ ⑤ $a^5 = a$ ⑥ $a^6 = a$

Clearly, a Only generates a, so it is not a generator

Start with element b:

① b ② $b^2 = b \cdot b = c$ ③ $b^3 = b \cdot c = d$ ④ $b^4 = b \cdot d = e$

⑤ $b^5 = b \cdot e = f$ ⑥ $b^6 = b \cdot f = a$

Since b generates all elements a, b, c, d, e, f. b is a generator of group. The group (G, *) is Cyclic with b as a generator.

(repeated)

(b) State and prove Lagrange's theorem.

(c) If G be a group with subgroup H & K. If $|G| = 660$, $|K| = 66$ & $K \subset H \subset G$ & find the possible value for $|H|$.

→ $|G| = 660$, $|K| = 66$, $K \subseteq H \subseteq G$ the Order of H must divide both $|G|$ & $|K|$ by Lagrange's theorem.

$|H|$ must divide $|G|$ (660) and must Contain $|K|$ (66)

$|H|$ should be Common divisors of 660 & 66.

gcd (660, 66) = 66

∴ The possible values of $|H|$ are divisors of 66.

∴ The possible values for $|H|$ are 1, 2, 3, 6, 11, 22, 33, 66. These are divisors of both $|G|$ & $|K|$ that

Satisfy the Conditions $K \subseteq H \subseteq G$.